

Jack Mislinski

Physiologist who builds and deploys AI software · Founder, Aevox · MSc Applied Physiology · NYU Computer Science · ex-Barclays
Brooklyn, NY · jackmis610@gmail.com · (312) 914-1773 · jackmislinski.com · github.com/jackmislinski · jackmislinski.substack.com

SUMMARY

A physiologist who builds and deploys the software himself. Founder of Aevox, a CPET and metabolic-testing platform deployed one isolated instance per lab: five sites live, from a CommonSpirit Health / EXOS performance center to independent labs, each on its own cart vendor and export format. The loop at every site: sit with the scientist, run the test, ingest their data as it actually arrives, find where the analysis fails, ship the fix that week. Graduate research in mitochondrial bioenergetics, including a multi-omics analysis of the MoTrPAC endurance-training dataset. NYU Computer Science; three years trading investment-grade credit at Barclays.

EXPERIENCE

Aevox — Founder & sole engineer

2024 – Present

- **Deployment.** One isolated deployment per clinic (own Vercel project, Clerk instance, Neon Postgres in the clinic's region), provisioned by a single script with per-clinic branding, DNS, and a first-visit checklist. Five sites live; the first is the Hybl Performance Center (CommonSpirit Health / EXOS).
- **Ingest.** Accepts a lab's export as uploaded: unit-aware column mapping, a Claude-proposed mapping checked against physics (Weir energy balance, VO_2/VCO_2 ranges) before it is accepted, and a per-sheet report that states exactly why a file was refused. Handles COSMED, MGC, PNOE, and VO_2 Master exports, lactate step sheets, and O_2 -only analyzers.
- **Engine.** Analysis engine at v13 with 500+ tests: ventilatory thresholds from deterministic detectors with the model as reviewer, not decider, and nothing committed until the clinician confirms; step-protocol detection from HR and VO_2 dips; lactate thresholds by first rise and Dmax ; grade-corrected flat-equivalent paces; running economy from steady-state stages only; reports that withhold what a missing input makes false and say why on page one.
- **Field loop.** Trial users' asks become shipped features in hours: a coach's per-stage treadmill grade became grade-corrected pace zones the morning he signed up; a lab's hand-graphed lactate sheet drove the step protocol within a day.
- **Go-to-market and paper.** Outbound to metabolic labs and performance centers (24% reply rate), discovery calls, pricing sheet, MSA, and order form. Set the HIPAA posture and wired the athlete consent record end to end.
- **Operations.** Run the company on scheduled Claude agents I built: they rank the day's commitments, draft outreach, and read upload and refusal metrics from every deployed instance. I keep the veto; the agents keep the calendar.

Barclays Investment Bank — Associate, Investment Grade Credit Trading

New York, NY · 2022 – 2025

- Market-made and managed risk on a **\$1B book** of front-end IG corporate bonds: daily two-way pricing and real-time coverage of 20+ buy-side accounts; built and owned the desk's analytics stack (Python, SQL, kdb/q) for PnL attribution, volumes, market share, and risk. **Tradeweb 2023 Top 5 Performer.**

RESEARCH

Mitochondrial oxidative capacity outpaces redox buffering during endurance training — manuscript in preparation

2026

Analysis of the MoTrPAC rat endurance-training dataset: 85 redox-relevant genes, 42,770 observations across 19 tissues, 5 omic layers, and 4 training time points. Redox buffering scales sub-linearly with oxidative capacity. Pipeline and draft public on GitHub.

SELECTED PROJECTS

Parsel — Healthcare × AI hackathon, winning entry

2026

A voice digital biomarker for Parkinson's, privacy-preserving: a self-referenced baseline and drift, never a diagnosis. Built the clinician dashboard and patient app from scratch (React, TypeScript; spoken prompts through an ElevenLabs proxy with browser-TTS fallback; MediaRecorder capture) on a FastAPI + Baseten backend. Live at parsel.health.

bioML — substrate utilization from wearable signals

2025

Multitask gradient-boosted model predicting VO_2 , RER, and substrate oxidation from PPG heart rate, power, CGM, and demographics. Subject-wise 5-fold CV ($n=100$): RER MAE 0.034, LT2 AUC 0.953, 38–59% better than the strongest baseline on VO_2 .

TECHNICAL SKILLS

AI / LLM: Claude API and Claude Code, tool use, adaptive thinking, cached prompts, scheduled agents, physics-checked proposals, evals.

Full-stack / infra: TypeScript, React 19, Vite, Next.js, Vercel serverless, Clerk, Neon Postgres, Cloudflare Workers, WebSockets, vitest.

ML / data: Python, PyTorch, LightGBM, scikit-learn, pandas, MLflow, SQL, kdb/q; time-series models on noisy biosignals, grouped CV.

Science: exercise and mitochondrial physiology, CPET, $\text{VO}_2/\text{VCO}_2/\text{RER}$, substrate oxidation, ventilatory and lactate thresholds, instrument validation, multi-omics (MoTrPAC), digital biomarkers.

EDUCATION AND ADDITIONAL

M.S. Applied Physiology, University of Colorado Colorado Springs · GPA 4.0 · 2025 – 2027. Mitochondrial bioenergetics and metabolic-cart validation (6.5% VO_2 bias vs. reference); two manuscripts in preparation.

B.A. Computer Science, New York University · 2020 – 2022. **United States Military Academy, West Point**, CS major · 2018 – 2020: Dean's List every semester, Cadet First Sergeant (top 2 of ~1,100).

Ultramarathoner (Tahoe Rim Trail 100, Run Rabbit Run 50) and Pacific Crest Trail thru-hiker (2,650 mi). Spanish and Portuguese.